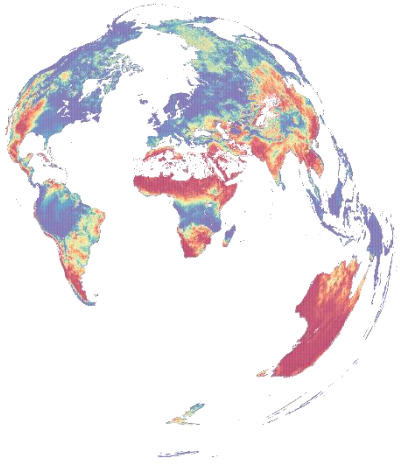
Geospatial Analysis with R



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Materials are available at the following address: https://fredi-agbohessou.github.io/courses/course1/geospatialR.rar

Course content

- **Section 0**: Preparation and prerequisites
- **Section 1**: Review of geographic information systems (GIS) and geospatial data
- **Section 2**: Handling vector data
- **Section 3**: Handling raster data
- **Section 4**: Interpolating raster data from point observations
- **Section 5**: How to manage a NetCDF file?
- **Section 6**: Advanced vector data manipulation

Section 0: Preparation and Prerequisites

What is this course about?

- Learn to use the R language to extract, manipulate, visualize, and analyze geospatial data from global databases.

Why This Course? (1)

🚀 **Most courses focus on tools, not skills**: Many spatial data analysis courses revolve around software like QGIS & ArcGIS, but these tools have limitations when it comes to automation, scalability, and customization.

💡 **Programming unlocks true flexibility**: With languages like R, you can automate repetitive analyses, manage large datasets, combine spatial and environmental data, and build reproducible workflows for soil-crop modeling and so much more.

🌍 **Agriculture is becoming increasingly data-driven**: Modern agronomy generates huge amounts of spatial data from soil sampling, remote sensing, drones, weather stations, GPS, and yield monitors. This course helps you turn these data into useful information for understanding spatial variability and making better agronomic decisions.

Why This Course? (2)

🔑 **Learn skills that are useful for your future career**: Programming-based spatial analysis can help you work on real-world problems such as precision agriculture, soil fertility mapping, crop monitoring, climate-smart agriculture, yield prediction, and environmental assessment.

🎯 **The goal:** Not just to learn R or spatial analysis, but to learn how to use data to answer agronomic questions.

Why get interested in spatial data analysis? (1)

1- Real-World applications across industries (*Agriculture, Environmental science, Urban planning, Health & epidemiology...***):**

- **Boost agricultural efficiency**: Use yield mapping, soil health analysis, and precision farming to maximize productivity and reduce input waste (e.g., variable rate fertilization).
- **Optimize traffic & mobility**: Analyze real-time traffic patterns and design efficient road networks (e.g., reducing congestion in cities).
- **Mapping disease outbreaks**: COVID-19 spread...
- **Retail & business intelligence**: Site selection for new stores based on population density and customer behavior.
- ...

WHY GET INTERESTED IN SPATIAL DATA ANALYSIS? (2)

2- Powerful insights beyond traditional data analysis: Unlike traditional data (spreadsheets, databases), spatial data adds an extra layer of meaning: location.

3- Growing demand for GIS & spatial analytics skills: With the rise of smart cities, remote sensing, and AI-powered mapping, there's increasing demand for professionals who can work with GIS, satellite imagery, and spatial databases.

4- Opportunities in machine learning & AI: Spatial data is at the core of autonomous vehicles, drone mapping, and AI-driven geospatial predictions. If you're into data science, spatial analytics can add a unique and valuable skill set.

5- Interdisciplinary & Future-Proof: Whether you come from computer science, environmental science, business, or engineering, spatial analysis complements many fields and offers future-proof career opportunities.

...

What will you gain from this course?

- Understand the fundamentals of raster and vector data.
- Learn to import, manipulate, and export geospatial data in R.
- Perform spatial operations such as overlays, intersections, buffering, and clipping.
- Analyze and visualize geospatial data using R's powerful packages.
- Interpolate raster data from point observations.
- Learn how to manage a NetCDF file.
- Apply geospatial analysis techniques to real-world problems.

R vs. QGIS for Geospatial Analysis: Why Choose R?

- ☑ Use QGIS if:

- You prefer a visual, GUI-based approach.
- You need quick GIS tasks (basic spatial analysis, map creation).

- ☑ Use R if:

- You need automation, reproducibility, and large-scale processing.
- You are working with advanced spatial statistics and modeling.
- You want to integrate spatial analysis with data science and machine learning.

PREREQUISITES

- Prior experience with R can be helpful.
- A good understanding of basic statistics and data visualization is useful but not mandatory.

Material

- R and Rstudio

Section 1: Review of Geographic Information Systems (GIS) and Geospatial Data

What is a GIS?

- A Geographic Information System (GIS) is a technology used to create, manage, analyze, and map all types of data *(esri.com)*.

- It is a tool that collects, stores, processes, and analyzes geographic data. It allows users to visualize this data as layered overlays on a georeferenced map, making it easier to extract, sort, and process the information according to the user's needs.

- A GIS links data to a map and integrates location data (where things are) with various types of descriptive information (the condition or characteristics of things at that location) *(esri.com)*.

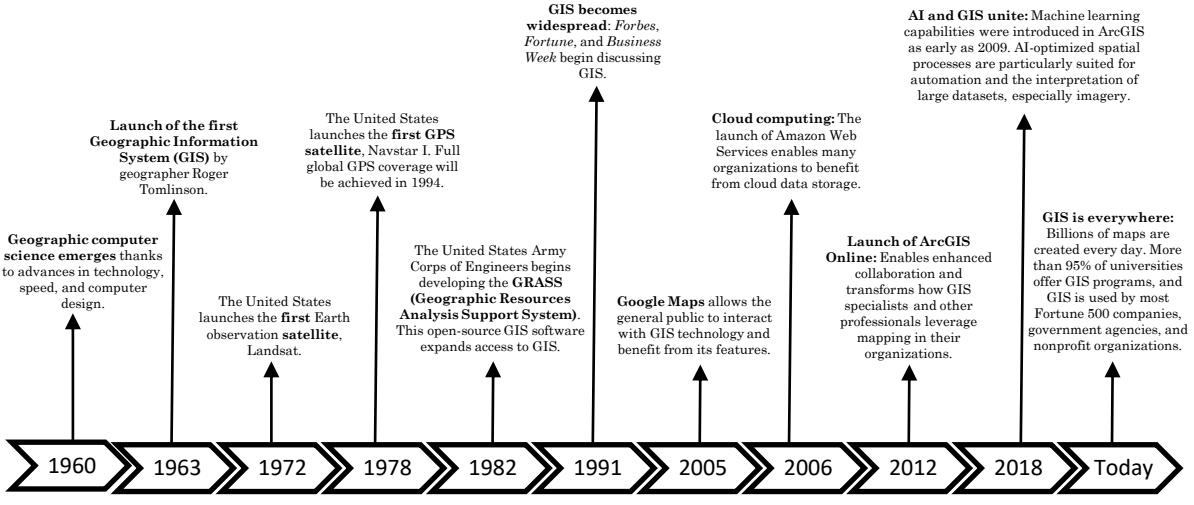


Examples of GIS with which you may already be familiar



R can also be used as a GIS

CHRONOLOGY OF THE DEVELOPMENT OF GIS: A FEW KEY DATES



What is the function of a GIS? More than just mapping (1)

- **Data management:** Storage, integration, and unification of data (same format).
- **Mapping and visualization:** Clarifying complex data, facilitating quick trend analysis, and improving decision-making by making information more accessible.
- **Spatial analysis:** Generating new insights.
- **Communication:** Conveying complex ideas more efficiently.

What is the function of a GIS? More than just mapping (2)

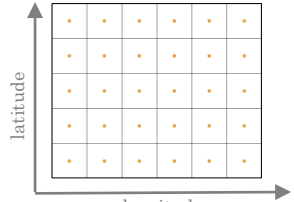


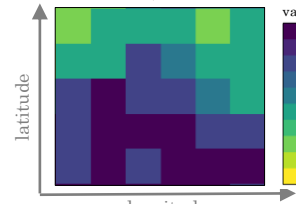
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4856	5555	3059	0	0	6	13476
5075	5767	3062	0	0	6	13910
4765	5358	2972	0	0	4	13099
11430	10795	5658	0	0	53	27936
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12035	10873	5873	0	0	71	28852
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11922	11214	5638	0	0	64	28838
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2632	3572	2290	0	0	11	8505
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file in xlsx format

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file in csv or txt format



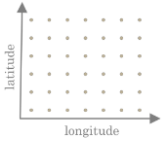


raster file in tiff or nc format

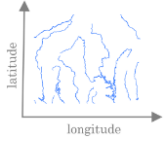
Main Components of Data Structure in a GIS: Spatial data (1)

Vector data

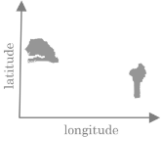
- **Points:** Used to represent geographic objects that can be located with a single coordinate (e.g., geodetic markers, trees).



- **Lines:** Represent geographic objects with a linear dimension (e.g., roads, rivers).



- **Polygons:** Represent closed surfaces (e.g., land parcels, lakes).



Main Components of Data Structure in a GIS: Spatial data (2)

Raster data (1)

- **Satellite images:** Captures of the Earth's surface taken from space.
- **Aerial Photographs:** Images taken from aircraft, often used for more detailed analysis.

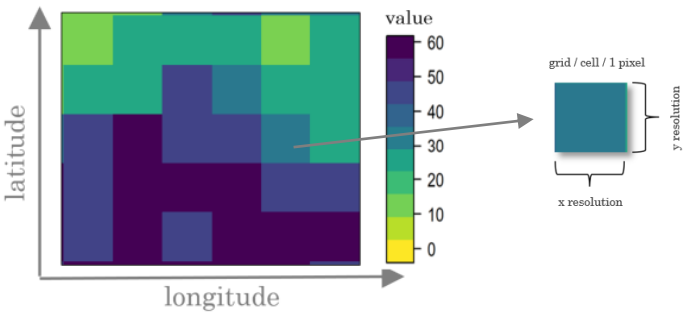


Google image of Cotonou (Benin) showing the Palais des Congrès, the Place de l'Amazone, and the new Ministerial City under construction.

Main Components of Data Structure in a GIS: Spatial data (3)

Raster data (2)

- **Raster image with georeferenced information**



Color indicates the value (of the data) assigned to the pixel (here, approximately 35).

The data can be "geolocated" using the x-y coordinate values.

Main components of data structure in a GIS: Attribute Data

Characteristics of geographic objects represented by spatial data. They are often stored in database tables and are linked to spatial data through unique identifiers (e.g., *fid: feature identifier*).

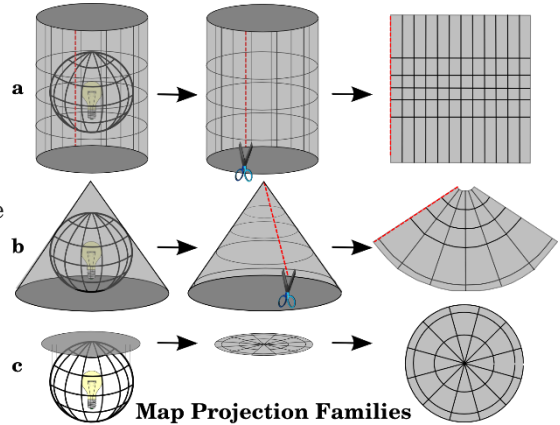
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184	184	-17,2301717	14,7	-17,1301717	14,6	144	11
185	185	-17,2301717	14,6	-17,1301717	14,5	144	4
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231	231	-17,130171700...	15	-17,0301717	14,9	144	5
232	232	-17,130171700...	14,9	-17,0301717	14,8	144	19

Main components of data structure in a GIS: Spatial Reference System

A GIS uses coordinate reference systems to project three-dimensional Earth locations onto a two-dimensional map.

A Spatial Reference System (SRS) defines how spatial data is projected and referenced on the Earth's surface. It includes coordinate systems, map projections, and geodetic datums to ensure accurate positioning and consistency across datasets.

For more information, see https://docs.qgis.org/3.4/en/docs/gentle_gis_introduction/

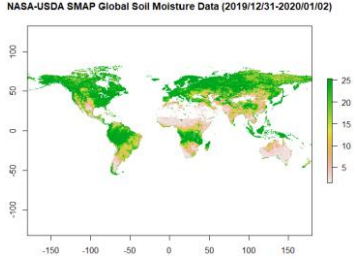


3 families of map projections: a) Cylindrical Projections; b) Conic Projections; c) Planar (or Azimuthal) Projections

Main components of data structure in a GIS: Metadata

Information that describes GIS data, including its origin, creation date, accuracy, and other contextual details necessary to understand and properly use the geographic data.

https://earth.gsfc.nasa.gov/hydro/data/nasa-usda-global-soil-moisture-data



NASA-USDA Enhanced SMAP Global Soil Moisture Data

The NASA-USDA Enhanced SMAP Global soil moisture data provides soil moisture information across the globe at 10-km spatial resolution. This data set includes: surface and subsurface soil moisture (mm), soil moisture profile (%), surface and subsurface soil moisture anomalies (-).The data set is generated by integrating satellite-derived Soil Moisture Active Passive (SMAP) soil moisture observations into the modified two-layer Palmer model using a 1-D Ensemble Kalman Filter (EnKF) data assimilation approach. The Palmer model is driven by precipitation and temperature observations provided by the United States (U.S.) Air Force 557th Weather Wing (formerly known as the U.S. Air Force Weather Agency, AFWA). The current SMAP data archive spans from March 31, 2015 and provides multiple soil moisture products developed using different algorithms. Here, we used the Level 3 (L3) SMAP product generated using the single channel algorithm (SCA) and distributed at a 9-km spatial resolution. Soil moisture anomalies were computed from the climatology of the day of interest. The climatology were estimated based on the full data record of the SMAP satellite observation and the 31 day centered moving window approach. The assimilation of the SMAP soil moisture observations help improve the model-based soil moisture predictions particularly over poorly instrumented areas of the world that lack good quality precipitation data. This dataset was developed by the Hydrological Science Laboratory at NASA's Goddard Space Flight Center in cooperation with USDA Foreign Agricultural Services and USDA Hydrology and Remote Sensing Lab.

Data availability:

01 April 2015 – present

3-days composites

Spatial coverage:

180°W – 180°E, 90°N-60°S

Data access and file formats:

NASA-USDA Enhanced SMAP Global Soil Moisture Data are available in both Gridded Binary (grb2), and Georeferenced Tagged Image File Format (GeoTiff). The binary format of the data can be accessed using the NASA Global Inventory Modeling and Mapping Studies (GIMMS) system. The tiff format of the data can be obtained using the Google Earth Engine.

Metadata in the form of a PDF or HTML file

Main components of data structure in a GIS: Data Formats

GIS data can be stored in various formats, each with its own advantages and disadvantages depending on the needs:

- **Shapefile (.shp):** A vector file format developed by ESRI.
- **GeoJSON:** A JSON-based format for encoding various geographic data structures.
- **KML (Keyhole Markup Language):** Primarily used with Google Earth.
- **TIFF/GeoTIFF:** A raster image format with georeferenced information.
- **NetCDF (Network Common Data Form) (.nc):** Used for storing multidimensional scientific data, often in climate and atmospheric sciences.

Set Up Your R Environment for Geospatial Analysis

- Installing R and Rstudio
- Creating an R project
- Installing packages for geospatial data analysis and visualization (raster, sf, ggplot, rgdal, etc.)

Section 2: Handling Vector Data

Section 3: Handling Raster Data

Section 4: Interpolating Raster Data from Point Observations

Section 5: How to Manage a NetCDF File?

Section 6: Advanced Vector Data Manipulation

EXERCISE

Exercise A: Vector Data Manipulation

Use shapefiles from the The Humanitarian Data Exchange database (https://data.humdata.org/m/dataset) to:

- Plot a map of African countries with a population between 10,000,000 and 20,000,000 (shapefile: geospatialR/shapefiles/africa_shp/Africa_Countries.shp).
- Represent, in both a graph and a map, the most frequent types of amenities in Benin and Nigeria (shapefile: hotosm_ben_points_of_interest_points_shp.shp or hotosm_nga_points_of_interest_points_shp.shp).
- Create a map showing areas more than 10 km away from the most frequent types of amenities in Benin or Nigeria (shapefile adm1 in point_of_interest/Nigeria or Benin).

BONUS: Use the geospatial shapefile R/shapefiles/bio_adm2/BIO_ADM2.shp to create a map showing the 100 regions in Sahelian countries with the highest average biomass production in 2019 (or, to go further, the 100 regions with the highest average biomass production over the period 2000-2019).

Exercise B: Raster Data Manipulation

Use the raster geospatialR/rasters/biomass_shl/BiomassValue2019.tif from the The Humanitarian Data Exchange database (https://data.humdata.org/m/dataset) to:

- Create a map showing annual biomass production in 2019 in Sahelian countries.
- Display country boundaries and names of Sahelian countries on the map.
- Show the average biomass production per country on the map for 2019.
- Extract a region from Senegal or Nigeria (1st map), increase the resolution (2nd map), and then decrease the resolution (3rd map).

Exercise B: Raster Data Manipulation

BONUS:

- Create an NDVI RasterBrick using the raster files in the folder geospatialR/rasters/NDVI/ with lapply or a for loop.
- Select and display only the first four layers.
- Calculate the mean, maximum, and minimum NDVI values for the first four layers.
- Plot a time series of NDVI (using all layers, not just the first four) at a point located in your home region, specifying the coordinates of the point (e.g., x = 15.5, y = 14; where x = longitude and y = latitude in UTM).

Note:

The NDVI (Normalized Difference Vegetation Index) is a remote sensing indicator used to measure "green vegetation" or plant health. It is calculated based on how plants absorb light in specific wavelengths, particularly in the visible and near-infrared spectrum. Values close to 1 indicate dense and healthy vegetation.Values close to 0 represent very low or no vegetation cover (such as bare soil or water).

Thank you for your attention

References

- *https://www.esri.com/fr-fr/what-is-gis/overview*
- *https://bookdown.org/nicohahn/making_maps_with_r5/docs/introduction.html*
- *https://data.humdata.org/*
- *https://hub.arcgis.com/search?collection=Dataset&type=Shapefile*